

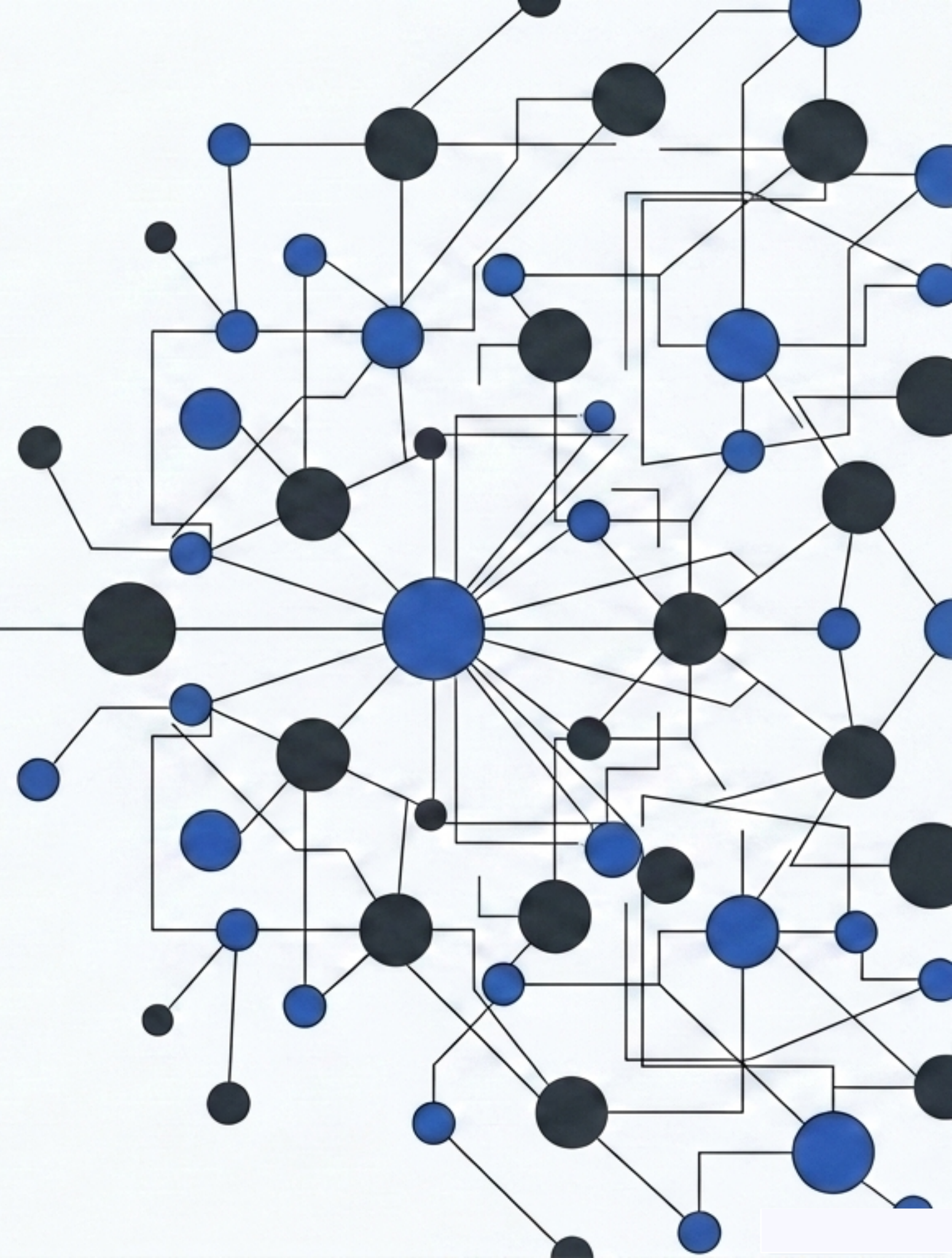
FinIntel AI

Graph-Aware Financial Intelligence
for Cybercrime Investigation

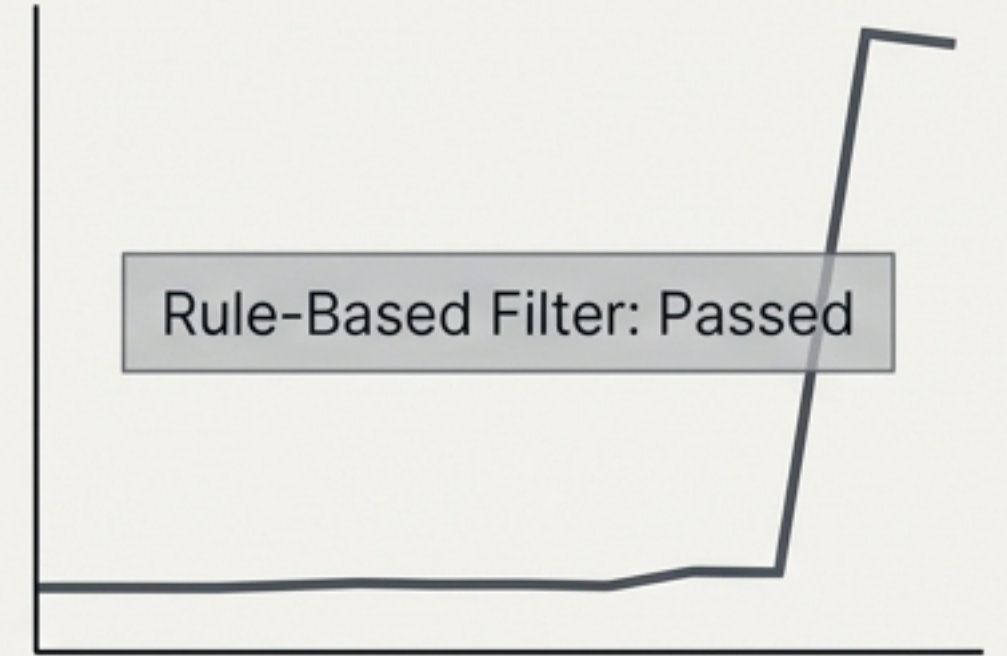
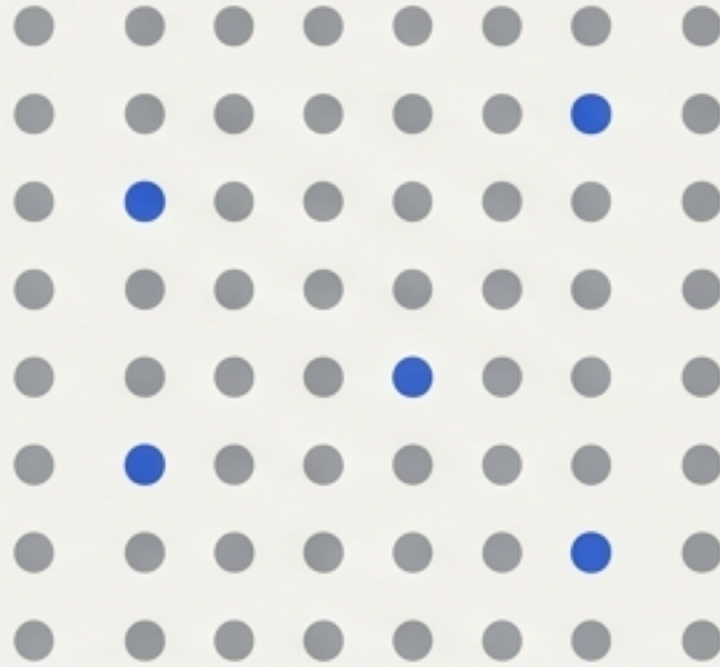
CIDECODE Hackathon 2026

CCITR – CID Karnataka | PES University, Bengaluru

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The Intelligence Gap: A Critical Operational Failure



Unstructured Chaos

Investigators manually sift through diverse inconsistent bank statement formats. The data ingestion process is entirely fragmented.

Siloed Transaction Data

Raw narrations lack structure. Data remains siloed making it impossible to query connections across thousands of transactions or multiple banks.

Rigid Rule-Based Failure

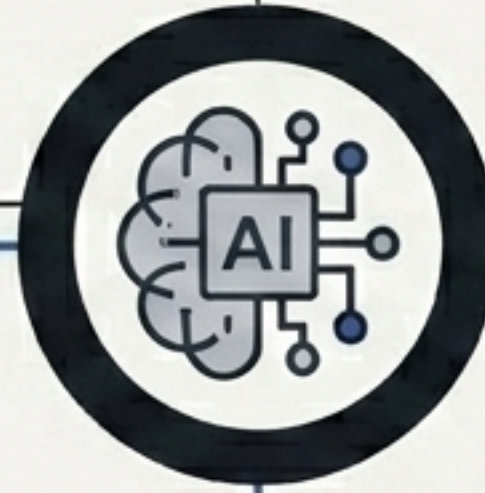
Existing systems rely on hard-coded static thresholds. Modern cybercriminals bypass these rules effortlessly using smurfing and mule networks.

The Core Insight: Money Laundering is Network Behavior, Not an Isolated Transaction.

Dimension	Traditional AML Systems	FinIntel AI
Primary Focus	Analyzes isolated individual transactions.	Analyzes hidden relationships and multi-hop fund flows.
Data Structure	Flat tabular databases unable to trace long chains.	Native Graph Database tracking entity-to-entity connections.
Threat Detection	Rigid static rules based on large volume alerts.	Hybrid Behavioral AI to catch smurfing and rapid movement .
Investigation Output	A generic dashboard of metrics and raw CSV exports.	Court-admissible AI-summarized evidence reports.

FinIntel AI Vision

Transforming raw multi-format bank statements into actionable court-ready investigation intelligence.

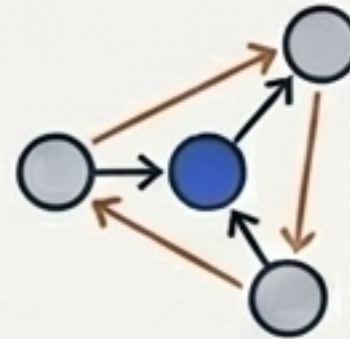


EXTRACT (Standardization)



Automated parsing of PDFs, Images, and CSVs.
NLP-driven normalization of chaotic bank narrations into a clean, searchable schema.

CONNECT (Graph Intelligence)



Multi-hop fund flow tracing.

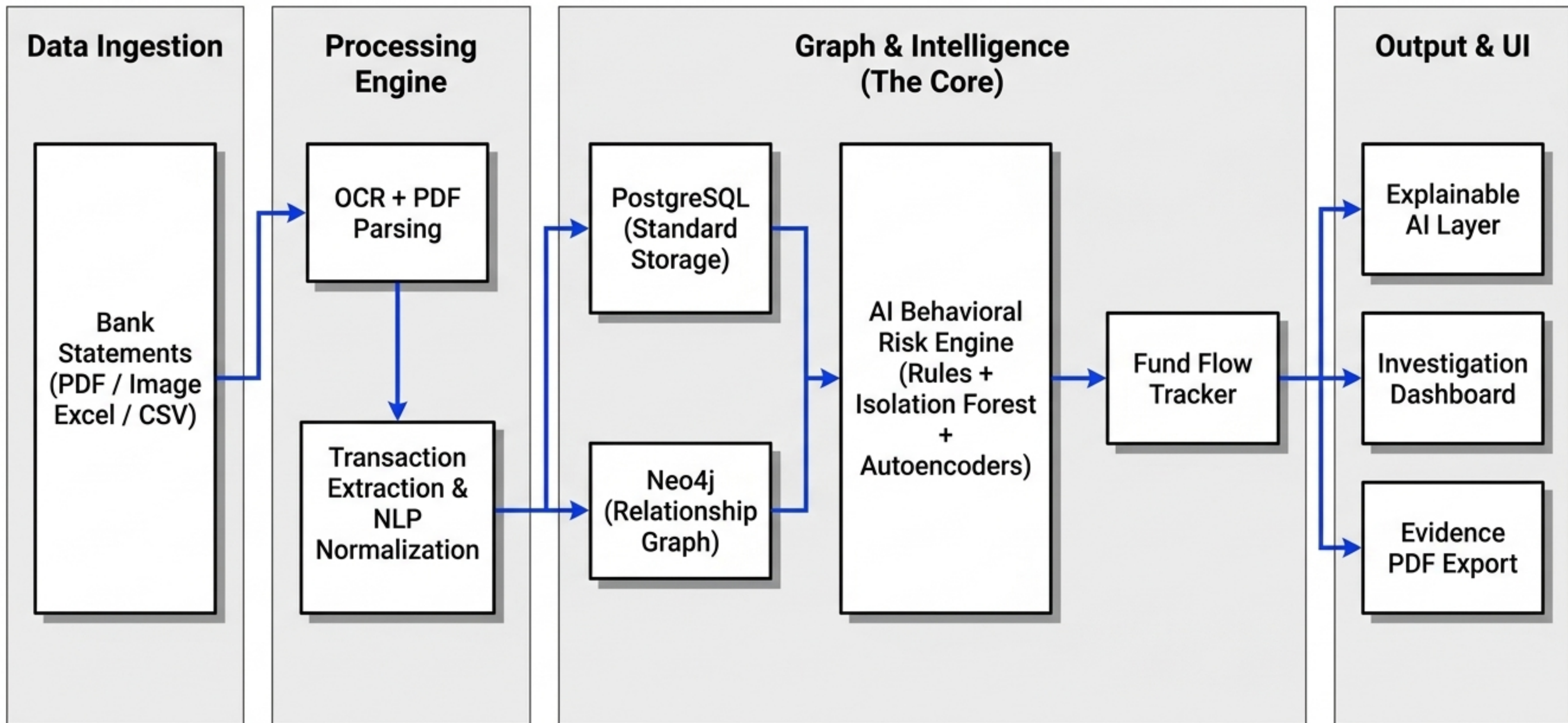
Uncovering hidden entity relationships, circular transactions, and mule networks across multiple accounts.

EXPLAIN (Actionable Evidence)



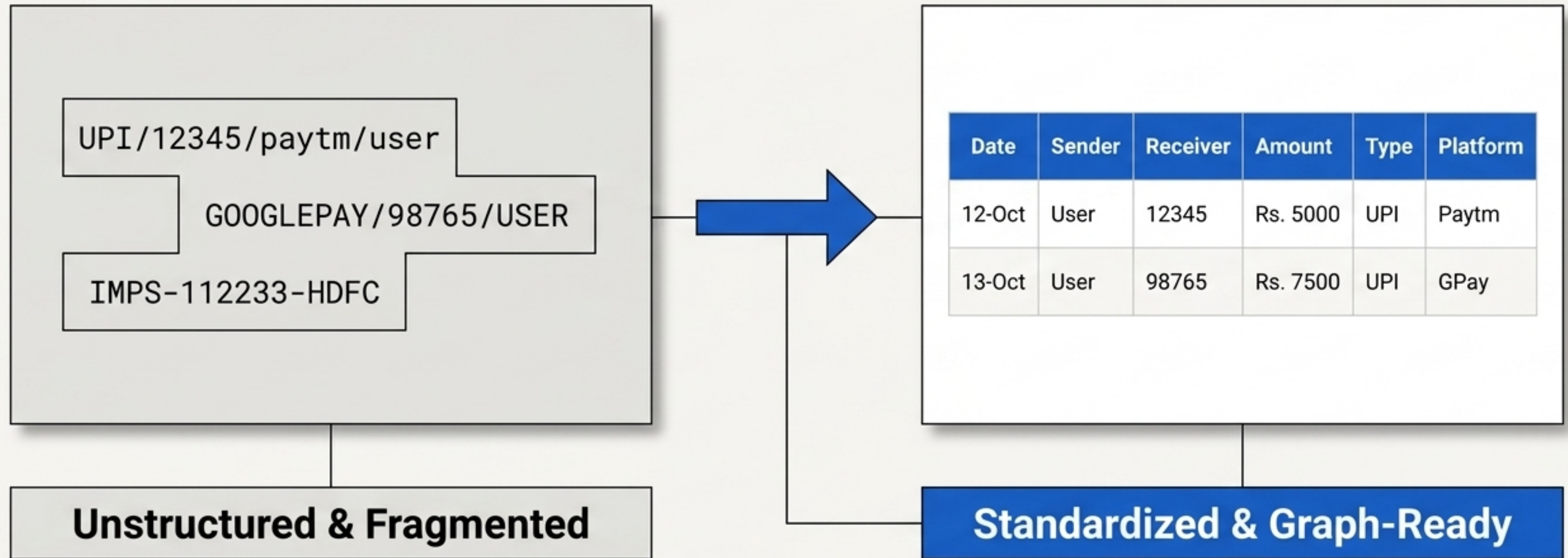
AI-generated reasoning that explains exactly why a cluster is suspicious. One-click export to court-admissible evidence reports.

End-to-End Intelligence Architecture



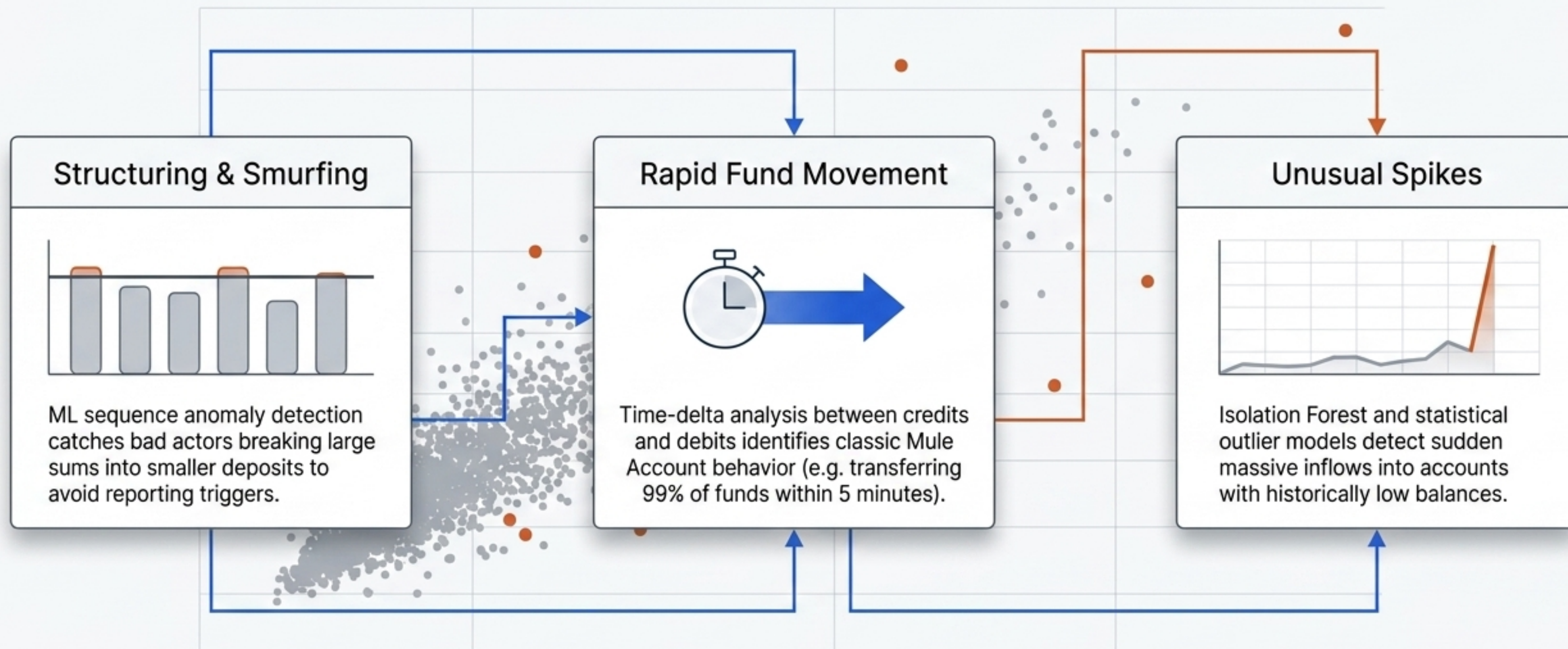
The Hidden Core: Parsing & Transaction Standardization

Without standardization AI-based pattern detection is useless. Our OCR + NLP pipeline normalizes chaotic inputs into a highly searchable database schema.



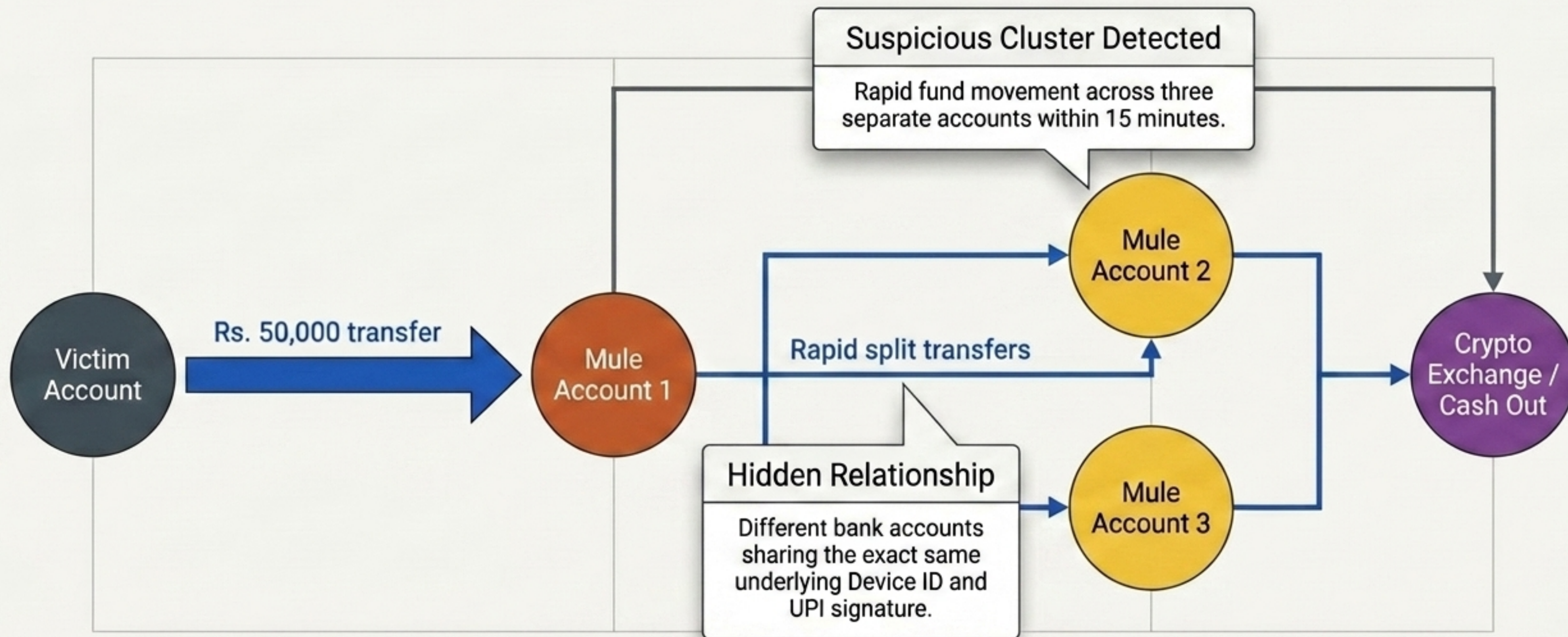
Hybrid AI Behavioral Risk Engine

Combining hard-coded Rule Engines with Machine Learning to detect sophisticated financial crime.



Financial Relationship Graph & Fund Flow Intelligence

FinIntel AI utilizes native graph databases to trace multi-hop laundering chains and circular transactions in milliseconds.



Explainable AI & Investigation Dashboard



The Intelligence Tech Stack

Frontend & Visualization

NEXT.js

tailwindcss



High-performance responsive UI and dynamic graph rendering.

Backend Architecture



Python



FastAPI



Rust API (Axum)



Asynchronous microservices for rapid data ingestion and API routing.

ML & Data Parsing



PaddleOCR



Tesseract



Multi-format OCR NLP normalization and hybrid anomaly detection models.

Database & Graph Layer



neo4j

Relational storage for transactions + Native graph engine for fund flow tracing.

Operational Impact & Evidence-Ready Reporting

The Impact

Days → Minutes

Automated parsing and normalization eliminates hundreds of hours of manual spreadsheet formatting.

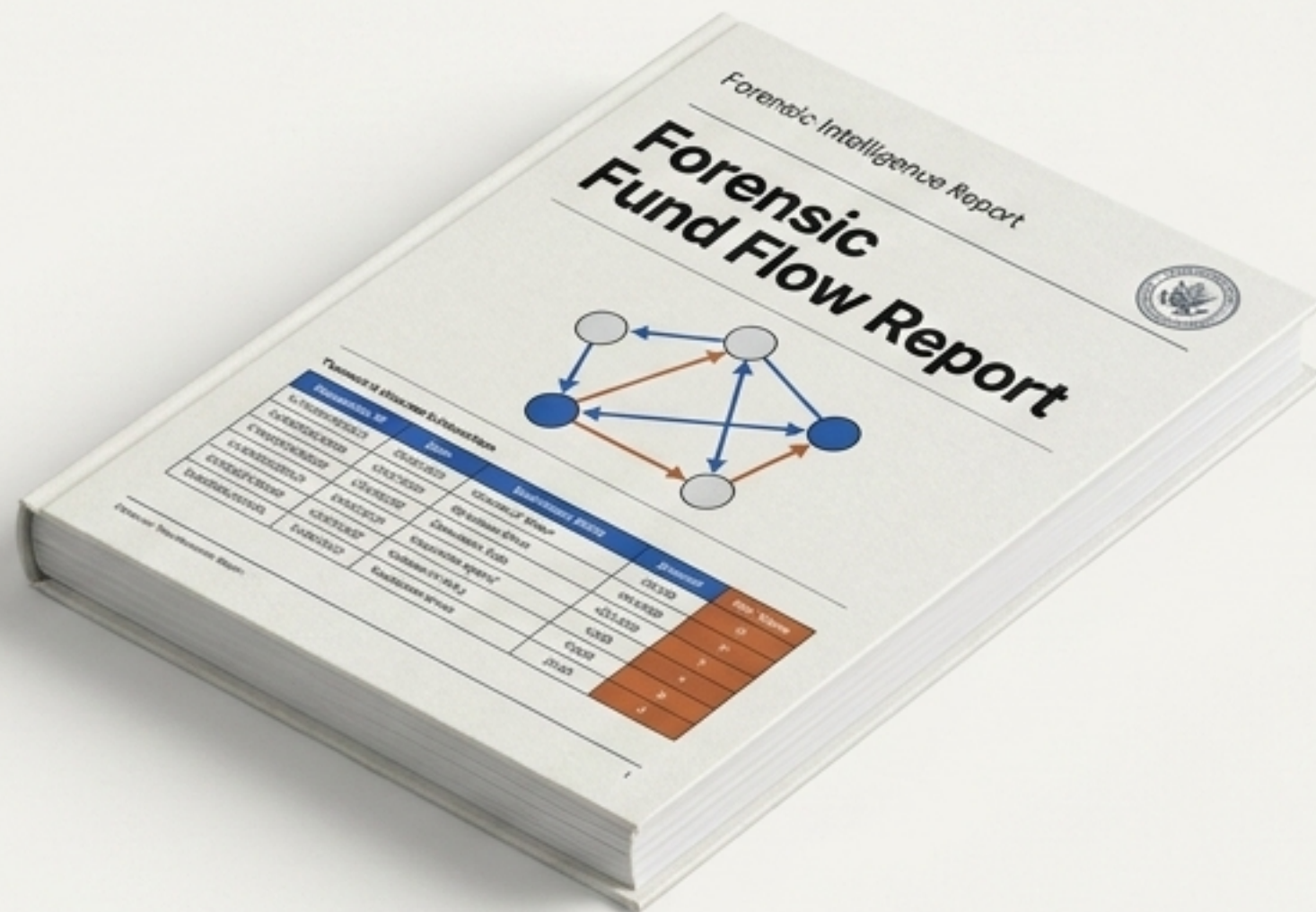
Network Visibility

Graph intelligence surfaces hidden connections and circular loops impossible to find in tabular data.

Defensible AI

Hybrid ML provides mathematically sound outlier detection without black-box confusion.

The Output



Court-Admissible Evidence Export

Compiles investigation summaries, AI reasoning, and fund flow timelines into PDFs ready for immediate submission by Cybercrime Units.